INCH-POUND

MIL-C-85049/39C
19 December 1989
SUPERSEDING
MIL-C-0085049/39B(USAF)
26 January 1987 and
MIL-C-85049/39A
4 March 1983

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, STRAIN RELIEF, 90°, SELF-LOCKING AND NONSELF-LOCKING, CATEGORY 4C (FOR MIL-C-38999 SERIES III AND IV CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the accessories described herein shall consist of this specification and the latest issue of MIL-C-85049.

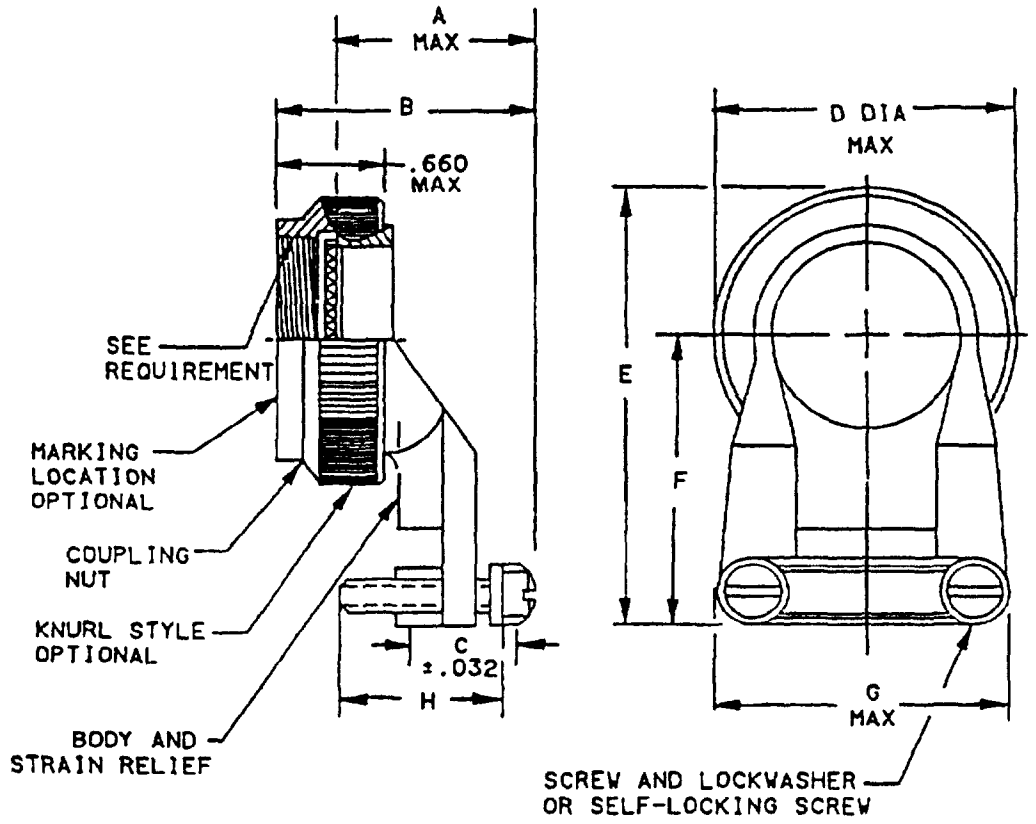


FIGURE 1. Dimension and configuration (self-locking only)

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Shell sizes and dimensions

Connector shell size	Shell size code (REF)	A max	B max overall length	C wire bundle accommodation range		D max dia	E max	F max size	G max	H screw length	Screw size
				Min	Max						
9	A	1.01 (25.7)	1.20 (30.5)	.098 (2 49)	.234 (5.94)	.658 (21.79)	1.38 (35.1)	1.00 (25.4)	.65 (21.6)	.500 (12.70)	6-32
11	B	1.01 (25.7)	1.20 (30.5)	.153 (3.87)	.234 (5.94)	.984 (24.99)	1.53 (38.9)	1.10 (27.9)	.90 (22.9)	.500 (12.70)	6-32
13	C	1 21 (30.7)	1.40 (35.6)	.190 (4.83)	.328 (8.33)	1.157 (29.39)	1 60 (40 6)	1.10 (27.9)	1.10 (27.9)	.625 (15.88)	6-32
15	D	1 26 (32.0)	1.45 (36 8)	260 (6 60)	.457 (11.61)	1 279 (32.49)	1.80 (45.7)	1.25 (31.6)	1.15 (29 2)	.750 (19.05)	6-32
17	E	1.41 (35.8)	1.60 (40.6)	.283 (7 19)	.614 (15.60)	1.406 (35 71)	1.93 (49 0)	1.30 (33.0)	1.30 (33.0)	.750 (19.05)	6-32

FIGURE 1 Dimensions and configurations (self-locking only) - Continued.

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Shell sizes and dimensions

Connector shell size	Shell size code (REF)	A max	B max overall length	C wire bundle accommodation range		D max dia	E max	F max size	G max	H screw length	Screw size
				Min	Max						
19	F	1 51 (38.4)	1.70 (43.2)	.325 (8.26)	.634 (16.10)	1.516 (38.50)	2.05 (52.1)	1.35 (34.3)	1.50 (38.1)	.750 (19.05)	8-32
21	G	1.66 (42.2)	1.85 (47.0)	.343 (8.71)	.698 (17.73)	1.642 (41.70)	2.35 (59.7)	1.60 (40.6)	1.60 (40.6)	.875 (22.23)	8-32
23	H	1.76 (44.7)	1.95 (49.5)	.381 (9.68)	.823 (20.90)	1.768 (44.91)	2.58 (65.5)	1.75 (44.5)	1.70 (43.2)	1.000 (25.40)	8-32
25	J	1.91 (48.5)	2.10 (53.3)	.418 (10.62)	.853 (21.66)	1.889 (47.98)	2.73 (69.3)	1.85 (49.5)	1.80 (45.7)	1.125 (28.58)	8-32

NOTES:
1. Dimensions are in inches.
2. Dimensions apply after plating.
3. Metric equivalents are given for general information only
4. Millimeters are in parentheses.

FIGURE 1. Dimensions and configurations (self-locking only) - Continued.

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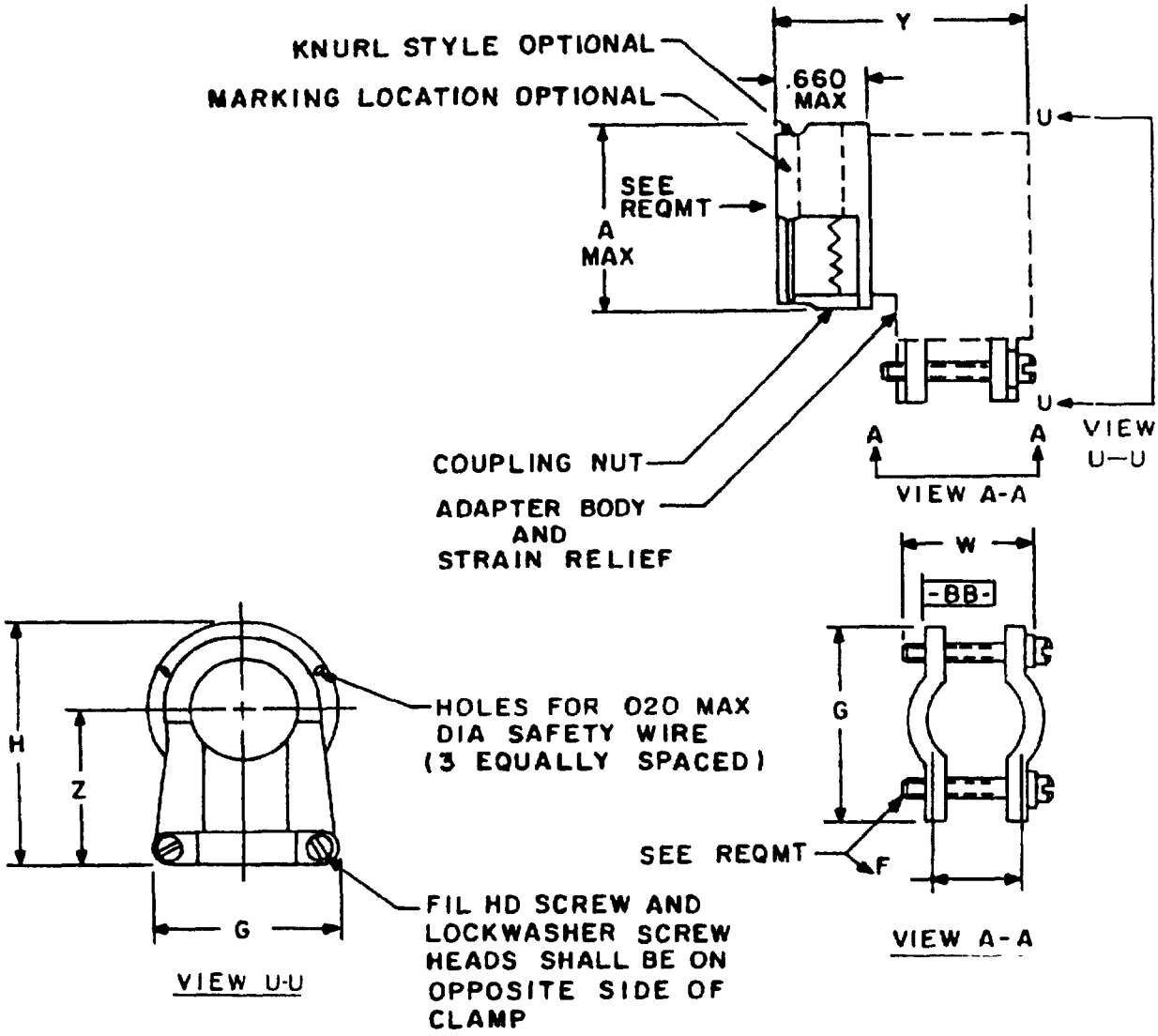


FIGURE 2. Dimensions and configurations (nonself-locking only).

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Shell size	A max dia of coupling ring	F wire bundle accommodation range		G max	Screw size	Y max	Z max	W ±.032 screw length	H max
		Min	Max						
9	.75 (19.1)	.098 (2.49)	.234 (5.94)	.85 (21.6)	6-32	1.20 (30.5)	1.00 (25.4)	.500 (12.70)	1.38 (35.1)
11	.85 (21.6)	.153 (3.87)	.234 (5.94)	.90 (22.9)	6-32	1.20 (30.5)	1.10 (27.9)	.500 (12.70)	1.53 (38.9)
13	1.00 (25.4)	.190 (4.83)	.328 (8.33)	1.10 (27.9)	6-32	1.40 (35.6)	1.10 (27.9)	.625 (15.88)	1.60 (40.6)
15	1.10 (27.9)	.260 (6.60)	.457 (11.61)	1.15 (29.2)	6-32	1.45 (36.8)	1.25 (31.8)	.750 (19.05)	1.80 (45.7)
17	1.25 (31.8)	.283 (7.19)	.614 (15.60)	1.30 (33.0)	6-32	1.60 (40.6)	1.30 (33.02)	.750 (19.05)	1.93 (49.0)
19	1.40 (35.6)	.325 (8.26)	.634 (16.10)	1.50 (38.1)	8-32	1.70 (43.2)	1.35 (34.3)	.750 (19.05)	2.05 (52.1)
21	1.50 (38.1)	.343 (8.71)	.698 (17.73)	1.60 (40.6)	8-32	1.85 (47.0)	1.60 (40.6)	.875 (22.23)	2.35 (59.7)
23	1.65 (41.9)	.381 (9.68)	.823 (20.90)	1.70 (43.2)	8-32	1.95 (49.5)	1.75 (44.5)	1.000 (25.40)	2.58 (65.5)
25	1.75 (44.5)	.418 (10.62)	.853 (21.67)	1.80 (45.7)	8-32	2.10 (53.3)	1.85 (47.0)	1.125 (28.58)	2.73 (69.3)

NOTES:
1. Dimensions are in inches.
2. Dimensions apply after plating.
3. Metric equivalents are given for general information only.
4. Millimeters are in parentheses.

FIGURE 2. Dimensions and configurations (nonself-locking only) - Continued.

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REQUIREMENTS:

Self-locking and nonself-locking styles:

Design and construction:

Dimensions and configuration: See figures 1 and 2.

Interface dimension shall conform to MIL-C-38999 series III and IV.

Accessories consist of a coupling nut and follower. The coupling nut shall be captivated to, and free to rotate on the follower. The accessories will not accommodate connectors using size 8, 4, 0 contacts. Figure 2 dimension Y and F_{MAX} are applicable when the end of the screw is flush with BB or a maximum of two screw threads protrude.

Clamps shall have no protrusions or sharp edges which may pinch cable.

Material and finish: See table I. Accessories shall be as shown in table I.

Clamp screws and lockwashers - corrosion-resistant steel, passivate.

TABLE I. Material and finish

Figure	Material	Finish
1	Aluminum alloy in accordance with MIL-C-85049	1/ N 2/ W
1	Corrosion resistant steel 300 series in accordance with QQ-S-763	S
2	Aluminum alloy in accordance with MIL-C-85049	A, W

1/ Not for Navy use. Air Force use is for space applications only.
2/ W is not for use in space application.

Self-locking styles only:

Self locking device within the coupling nut shall be corrosion resistant material and shall provide a positive detent. Couplings with self-locking devices shall meet all the performance requirements specified herein for its specific category and the following additional tests.

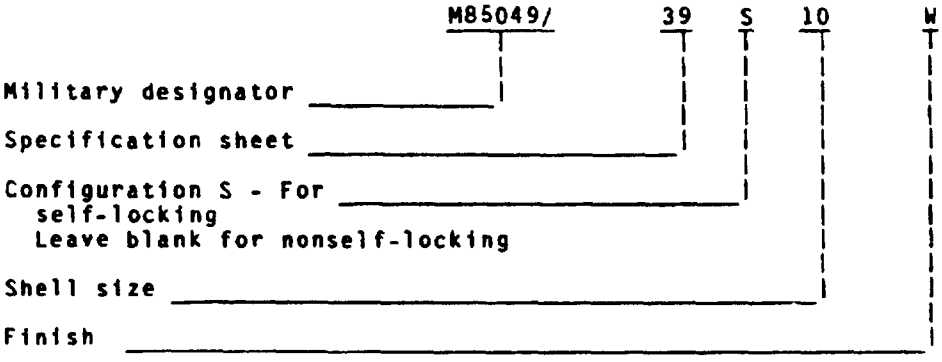
Life cycle: The coupling with the locking device engaged shall be rotated sixty full turns in a clockwise direction and sixty full turns counter clockwise, prior to standard qualification testing. Cycle speed shall not exceed one full turn per second.

Locking: Coupling nut torque shall be 80 percent of coupling thread strength values given for medium and light duty accessories as specified in table III of MIL-C-85049. In table III when N/A appears, the values shall be 80 percent of those values used in qualification testing for coupling thread strength. Tolerance shall be ±5 inch pounds.

Vibration. Backshells shall be subjected to test parameters of applicable connector specifications. The coupling torque shall be within +20, -10 inch pounds of the initial value.

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Part number: See example below:



QUALIFICATION:

See MIL-C-85049, category 4B.

QPL evaluating activity. Naval Avionics Center, Code B/444, 6000 East 21st Street, Indianapolis, Indiana 46219-2189.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force 11, 15, 19, 99

User activity:
Army - AV

Civil agencies coordinating activity:
NASA - NA

Preparing activity:
Air Force - 85

Agent.
DLA - ES

(Project 5935-3686-02)

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